

FERMAT'S LITTLE THEOREM

If p is a prime number and a is an integer not divisible by p , then $a^{p-1} \equiv 1 \pmod{p}$

Equivalently $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$

because if $p|a$, then a^p & a are both $0 \pmod{p}$

Proof

1. PROOF BY MATHEMATICAL INDUCTION:-

Base case: $a=0$

$\Rightarrow 0^p \equiv 0 \pmod{p}$ which holds trivially.

Indy $a=1$

$\Rightarrow 1^p \equiv 1 \pmod{p}$

Inductive hypothesis:

For some $k \geq 1$, we have $k^p \equiv k \pmod{p}$

Induction proof:

Now, we have to show that: $(k+1)^p \equiv (k+1) \pmod{p}$

Using binomial theorem, we expand,

$$(k+1)^p = \sum_{i=0}^p {}^p C_i k^{p-i} 1^i$$

For prime p , every binomial coefficient $\binom{p}{i} = \frac{p!}{i!(p-i)!}$

with $1 \leq i \leq p-1$ is divisible by p .

[$\because p$ is in the numerator $p!$]

$\therefore \pmod{p}$ all the intermediate terms vanish.

$$\therefore (k+1)^p \equiv \binom{p}{0} k^p + \binom{p}{p} 1^p \equiv k^p + 1 \pmod{p}$$

$$[\because \binom{p}{0} = \binom{p}{p} = 1]$$

By induction hypothesis $k^p \equiv k \pmod{p}$ and base case $1 \equiv 1 \pmod{p}$, we get

$$k^p + 1 \pmod{p} \equiv k + 1 \pmod{p}$$

$$\therefore (k+1)^p \equiv (k+1) \pmod{p}$$

This completes the inductive step.

$\therefore$ By induction, $a^p \equiv a \pmod{p} \forall a \geq 0$

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2. PROOF USING BINOMIAL COEFFICIENTS :-

For any $k \in \mathbb{Z}$, consider $k^p - (k-1)^p$.

By binomial expansion, we get,

$$k^p = [(k-1) + 1]^p = \sum_{i=0}^p {}^p C_i (k-1)^{p-i} 1^i$$

we know that,

For prime p , every binomial coefficient ${}^p C_i = \frac{p!}{i!(p-i)!}$

with $1 \leq i \leq p-1$ is divisible by p .

[$\because p$ is in the numerator $p!$]

Thus, $k^p \equiv {}^p C_0 (k-1)^p + {}^p C_p = (k-1)^p + 1 \pmod{p}$

$\Rightarrow k^p - (k-1)^p \equiv 1 \pmod{p}$ for each $1 \leq k \leq a$

Now, summing this congruence for $k=1$ upto $k=a$, we get,

$$\sum_{k=1}^a (k^p - (k-1)^p) = a^p - 0^p = a^p$$

$$\text{and } \sum_{k=1}^a 1 = a$$

$$\therefore a^p \equiv a \pmod{p} //$$

3. PROOF USING LAGRANGE'S THEOREM :-

Consider the multiplicative group of integers modulo p :

$$G = \{1, 2, \dots, p-1\} \text{ with multiplication mod } p.$$

By Bezout's identity, every $1 \leq a < p$ has a modular inverse.

For any a ,

let k be the smallest positive integer such that

$$a^k \equiv 1 \pmod{p} \text{ where } k \text{ is the order of } a \text{ in } G.$$

Lagrange's Theorem states that the order of an element must divide the order of the group.

$$\therefore k \mid |G| = p-1$$

So, we write $p-1 = k \cdot m$ for some $m \in \mathbb{Z}$.

Then

$$a^{p-1} = a^{km} = (a^k)^m \equiv 1^m = 1 \pmod{p}$$

$$[\because a^k = 1 \text{ by def}]$$

$$\therefore a^{p-1} \equiv 1 \pmod{p} //$$

4. STANDARD NUMBER THEORETIC PROOF USING MODULAR ARITHMETIC & PERMUTATION:-

Let p be a prime and a be an integer with $p \nmid a$
meaning, $\gcd(a, p) = 1$.

Consider a set of integers $\{1, 2, 3, \dots, p-1\}$ which are all the non-zero residues mod p .

Multiplying each element of this set by a and reducing it to mod p , we get

$$X = \{a \bmod p, 2a \bmod p, \dots, (p-1)a \bmod p\}$$

Because a is invertible mod p , no two of these $p-1$ products are congruent mod p and none is $0 \bmod p$.

Also, if $ja \equiv ka \pmod{p}$ for some $1 \leq j < k \leq p-1$, then dividing both side by a would give $j \equiv k \pmod{p}$ which is impossible for $1 \leq j < k < p$.
valid since $a \nmid p$ are relatively prime.

Thus these $p-1$ values $a, 2a, \dots, (p-1)a \bmod p$ are all distinct and non-zero.

Meaning they are just a permutation of $1, 2, \dots, p-1$

Since $a, 2a, 3a, \dots, (p-1)a$ is the same set of residues as $1, 2, 3, \dots, p-1$, then we must have,

$$a \cdot 2a \cdot 3a \cdot \dots \cdot (p-1)a \equiv 1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-1) \pmod{p}$$

①

Consider LHS,

$$\begin{aligned} a \cdot 2a \cdot \dots \cdot (p-1)a &= a^{p-1} \cdot (1 \cdot 2 \cdot 3 \cdot \dots \cdot (p-1)) \\ &= a^{p-1} (p-1)! \end{aligned}$$

$$\therefore \textcircled{1} \Rightarrow a^{p-1} (p-1)! \equiv (p-1)! \pmod{p} \quad \text{---} \textcircled{2}$$

Since $(p-1)!$ is not divisible by p , i.e., $\gcd((p-1)!, p) = 1$, $(p-1)!$ has a multiplicative inverse mod p .

So we can cancel $(p-1)!$ from both sides.

So $\textcircled{2}$ becomes $a^{p-1} \equiv 1 \pmod{p}$ //